


CSEC 6 - Natural Language Processing

# Truncation Impact Score: A Diagnostic Metric for Quantifying Salient Content Loss in Long-Form Biomedical Summarization

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## Abstract

Truncation is a persistent and underdiagnosed challenge in long-document biomedical summarization. Dominant architectures such as BART are constrained to a 1,024-token input window, yet the average PubMed article spans roughly 2,700 tokens, meaning standard models discard $\sim$62% of raw source tokens before inference begins (median salience-weighted loss: 55.92%). Standard metrics—ROUGE and BERTScore—measure output quality but cannot isolate whether a performance gap stems from limited model reasoning or information that was never made available. This paper introduces the **Truncation Impact Score (TIS)**, a model-agnostic, pre-inference diagnostic metric that quantifies the salience-weighted fraction of reference-relevant content falling beyond a model's context window, computed via global TF-IDF sentence weighting and unigram overlap requiring no model forward pass. We compare three systems fine-tuned on 12,000 PubMed articles via LoRA on a single NVIDIA A5000 24GB GPU: BART-large (1,024-token baseline), LED-large (hardware-capped at 4,096 tokens), and BART-Oracle (TF-IDF sentence selection within 1,024 tokens). BART-large achieves the highest scores (ROUGE-1: 0.3657, ROUGE-2: 0.1318, ROUGE-L: 0.2056) across 1,500 test articles. Despite LED's larger context window, its ROUGE-2 (0.0798) lags BART by 65.2%, attributed to *positional lead bias* in PubMed abstracts. TIS-stratified analysis reveals a monotonic convergence: the BART–LED ROUGE-2 gap narrows from $-0.0631$ (Low TIS) to $-0.0454$ (High TIS), establishing TIS as a principled pre-inference signal for data-driven model selection.

# 1 Introduction

The exponential growth of biomedical literature presents a formidable barrier to evidence synthesis. The PubMed repository currently indexes over 40 million citations [16], and the volume of new publications continues to rise. Automatic abstractive summarization offers a scalable mechanism to assist researchers and clinicians in navigating this corpus. However, the dominant architectures for sequence-to-sequence generation—most notably BART [12]—operate under a severe architectural constraint: a 1,024-token input limit. Given that the average PubMed article in our corpus spans 2,729 tokens (train split), standard models silently discard approximately 62% of raw source tokens before a single output token is generated. Crucially, this is distinct from the salience-weighted loss: the Truncation Impact Score developed in this work measures the fraction of *reference-relevant* content lost, with a test-set median of 0.5592.

This truncation is not uniform in its consequences. Biomedical articles follow the IMRAD structure (Introduction, Methods, Results, Discussion), where the methodology and empirical findings—the most scientifically critical content for a summary—appear in the middle and later sections, well beyond the 1,024-token boundary for most papers. The resulting information loss is systematic and domain-specific, yet it is invisible to output-centric evaluation metrics such as ROUGE [13] and BERTScore [24], which assess the quality of what was generated rather than what was discarded.

Long-context architectures such as the Longformer Encoder-Decoder (LED) [3] provide an architectural remedy by extending the context window to 16,384 tokens via sparse attention. However, the field lacks a diagnostic mechanism to predict *when* this additional capacity is necessary and to what degree it compensates for the truncation bottleneck. This gap motivates the **Truncation Impact Score (TIS)**, a pre-inference, model-agnostic metric that quantifies salience-weighted content loss before any model inference is performed.

We compute TIS for each article as a diagnostic score, then train and evaluate three summarization systems using LoRA fine-tuning: a BART-large baseline, an LED-large long-context model (hardware-capped at 4,096 tokens), and a BART-Oracle intelligent truncation baseline. Our primary contributions are:

- The formal definition of TIS, a salience-weighted diagnostic metric computed from TF-IDF sentence weights and reference overlap, requiring no model inference.
- A controlled end-to-end pipeline comparing BART-large, LED-large, and BART-Oracle on 1,500 PubMed test articles, fine-tuned via LoRA on a single NVIDIA A5000 24GB GPU.
- Empirical evidence that TIS-stratified evaluation reveals a monotonic convergence of the BART–LED ROUGE-2 gap as truncation severity increases, validating TIS as a meaningful diagnostic signal.
- An analysis of the counter-intuitive finding that BART outperforms LED overall, attributed to positional lead bias in PubMed abstract authoring conventions.

# 2 Related Work

## 2.1 Biomedical NLP and the Evidence Synthesis Challenge

Chaves et al. [4] confirmed that biomedical text summarization poses distinct challenges from general-domain NLP: a specialized vocabulary, the IMRAD document structure distributing salient content across thousands of tokens, and clinical demands for factual precision over linguistic fluency [20]. Adams et al. [1] documented that models frequently fail to synthesize evidence across long clinical contexts, while Ateia et al. [2] showed that even state-of-the-art LLMs exhibit domain-specific hallucination patterns on biomedical benchmarks, underscoring the need for pre-inference diagnostic tools.

## 2.2 Long-Document Transformer Architectures

Standard Transformer self-attention has $O(L^2)$ complexity relative to sequence length $L$ [5], forcing BART's 1,024-token limit to discard the majority of a typical PubMed article. LED [3] addresses this with sliding-window local attention and global attention on the BOS token, achieving $O(L \cdot w)$

complexity. BigBird [22] adds random attention blocks; LongT5 [8] applies sparse attention to T5. SCROLLS [19] demonstrated that long-context models outperform truncated baselines in aggregate, but without document-level diagnostics to explain instance-level gaps—a gap TIS is designed to fill. Critically, LED-large is initialized directly from BART-large weights [3], sharing all 406 million parameters, so any observed performance differential is attributable to the attention mechanism and context window, not pre-training disparities.

Liu et al. [14] identified the "Lost in the Middle" phenomenon: models exhibit a U-shaped utilization curve where central content is systematically underutilized. This is important for our study—even when LED's context makes biomedical content *available*, positional underutilization may prevent effective generation from it.

### 2.3 Neural Abstractive Summarization

Large pre-trained models—BART [12], T5 [18], PEGASUS [23]—established fine-tuning as the standard summarization paradigm. However, neural abstractive systems introduce significant fidelity risks: Pagnoni et al. [17] taxonomized three categories of factual error (predicate, entity, and circumstance errors) that are especially consequential in biomedical contexts [20]. Koh et al. [10] documented that optimizing for ROUGE often occurs at the expense of factual consistency.

### 2.4 Parameter-Efficient Fine-Tuning

Low-Rank Adaptation (LoRA) [9] freezes pre-trained weights and injects trainable low-rank decomposition matrices $\Delta W = AB^T$, where $A$ is Gaussian-initialized and $B$ is zero-initialized. This reduces trainable parameters by orders of magnitude while maintaining competitive fine-tuning quality. The BioLaySumm experiments of Liu et al. [15] established rank 8 for BART and rank 4 for LED as a practical configuration for single-GPU scientific summarization—the same configuration adopted here.

### 2.5 Evaluation Metrics and Diagnostic Gaps

ROUGE [13] measures n-gram recall and precision; BERTScore [24] uses contextual embeddings for semantic similarity. Faithfulness metrics such as SummaC [11] and QAFactEval [7] verify factual grounding. Despite this progress, all these metrics are *output-centric*: they assess what was produced, not what the model failed to observe. They cannot distinguish reasoning failure from information deprivation. TIS fills this gap as a *pre-inference, source-centric* metric that quantifies salience-weighted content loss before any generation occurs.

## 3 Data & Resources

### 3.1 Dataset

We use the PubMed Article Summarization dataset [6], a publicly available collection of biomedical article-abstract pairs. We sample the following fixed subsets:

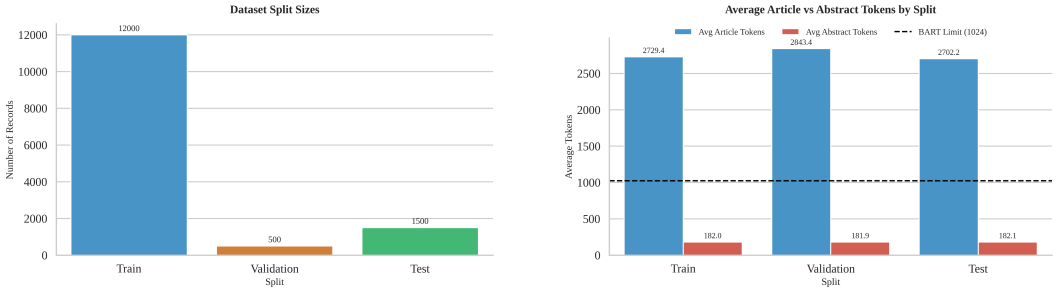
- **Training:** 12,000 article-abstract pairs for fine-tuning.
- **Validation:** 500 pairs for monitoring training dynamics.
- **Test:** 1,500 pairs reserved exclusively for final evaluation.

### 3.2 Corpus Statistics and Truncation Severity

Table 1 presents token-level statistics computed using the BART tokenizer. Across all splits, the mean article length ranges from 2,702 to 2,843 tokens—far exceeding BART's 1,024-token context limit. The average abstract, at roughly 182 tokens, fits comfortably within this window, confirming that the information bottleneck lies entirely on the source side. Figure 1 visualizes the dataset composition and token-length profiles.

**Table 1** – Corpus statistics for the sampled PubMed splits.

| Split | Records | Avg. Article Tokens | Avg. Abstract Tokens | Median Article Tokens |
|---|---|---|---|---|
| Train | 12,000 | 2,729.36 | 181.99 | 2,322.5 |
| Validation | 500 | 2,843.44 | 181.86 | 2,355.0 |
| Test | 1,500 | 2,702.22 | 182.10 | 2,344.0 |




**(1a)** Record counts per dataset split.

**(1b)** Average article vs. abstract token length by split.

**Figure 1** – Dataset composition and token-length profiles. Article lengths consistently exceed BART's 1,024-token context window across all splits, while abstract lengths remain well within the target generation budget.

### 3.3 Preprocessing

All texts were tokenized using HuggingFace tokenizers for BART-large (`facebook/bart-large-cnn`) and LED-large (`allenai/led-large-16384`). Source articles were truncated to each model's context limit: 1,024 tokens for BART and BART-Oracle, and 4,096 tokens for LED (hardware-capped from the theoretical 16,384-token maximum). Target summaries were capped at 256 tokens. Global attention was enabled on the beginning-of-sequence (BOS) token for LED inputs. No data augmentation was applied.

## 4 Methodology

### 4.1 System Architecture Overview

Figure 2 illustrates the end-to-end pipeline, spanning corpus ingestion, TIS computation, LoRA fine-tuning, and TIS-stratified evaluation.

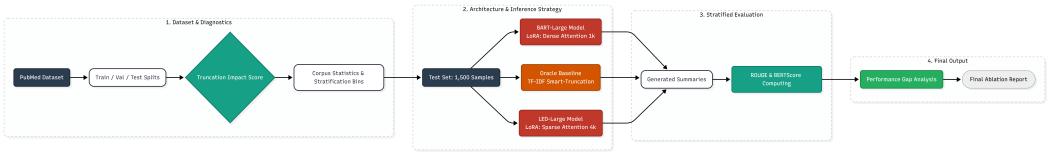



**Figure 2** – End-to-end pipeline architecture. The pipeline proceeds from corpus sampling through TIS computation (a pre-inference step requiring no model forward pass) to LoRA fine-tuning of BART and LED, followed by TIS-stratified evaluation of all three model systems.

### 4.2 Truncation Impact Score (TIS)

TIS is a model-agnostic, pre-inference metric that quantifies the salience-weighted fraction of reference-relevant content falling beyond a model's context window $C$. Given a source document $d$, a context limit $C$, and a reference abstract $r$, TIS is defined as:

$$\mathrm{TIS}(d,\, C,\, r) = \frac{\displaystyle\sum_{s_i \in d_{[C:]}} w(s_i) \cdot \mathrm{overlap}(s_i,\, r)}{\displaystyle\sum_{s_j \in d} w(s_j) \cdot \mathrm{overlap}(s_j,\, r)} \tag{1}$$

where $d_{[C:]}$ denotes the suffix of $d$ consisting of all sentences whose tokens begin beyond position $C$; $w(s_i)$ is the global TF-IDF importance weight of sentence $s_i$; and $\mathrm{overlap}(s_i, r)$ is the unigram recall of sentence $s_i$ against the reference abstract $r$.

**Sentence salience weight.** The TF-IDF weight $w(s_i)$ aggregates term-level importance across all tokens in sentence $s_i$:

$$w(s_i) = \frac{1}{|s_i|} \sum_{t \,\in\, s_i} \mathrm{tf}(t,\, d) \cdot \log \frac{N+1}{df(t)+1} \tag{2}$$

where $\mathrm{tf}(t, d)$ is the term frequency of token $t$ in document $d$; $N$ is the total number of training documents in the fitted corpus (5,000 articles); $df(t)$ is the document frequency of token $t$; and $|s_i|$ is the number of tokens in sentence $s_i$. Length normalization prevents longer sentences from being systematically over-weighted.

**Unigram overlap.** The overlap function measures the fraction of tokens in $s_i$ that also appear in the reference abstract $r$:

$$\mathrm{overlap}(s_i,\, r) = \frac{|\{t \mid t \in s_i\} \cap \{t \mid t \in r\}|}{|\{t \mid t \in s_i\}|} \tag{3}$$

A TIS of 0 indicates that all summary-salient content lies within the context window; a TIS of 1 means all salient content has been discarded. The joint weighting $w(s_i) \cdot \mathrm{overlap}(s_i, r)$ ensures that a sentence contributes to TIS only if it is both globally important (high IDF score) *and* locally relevant (overlaps with the reference), preventing high-IDF but irrelevant sentences from inflating the score.

### 4.3 TIS Difficulty Bins

To enable stratified evaluation, test articles are partitioned into three difficulty bins based on their computed TIS values (Table 2). The skew toward High TIS (56.5% of test articles) confirms that severe truncation is the dominant condition for PubMed documents under a 1,024-token limit, with a corpus-wide median TIS of 0.5592.

**Table 2** – TIS bin distribution across the 1,500-sample test set.

| Bin | TIS Range | Records | Share | Mean TIS |
|---|---|---|---|---|
| Low | $< 0.2$ | 349 | 23.3% | 0.0353 |
| Medium | $[0.2,\, 0.5)$ | 303 | 20.2% | 0.3581 |
| High | $\geq 0.5$ | 848 | 56.5% | 0.6966 |

### 4.4 Model Systems

We evaluate three model systems under identical preprocessing and evaluation conditions:

- **BART-large (1k Baseline):** Standard BART-large with right-side truncation to 1,024 source tokens. Serves as the primary reference baseline.
- **LED-large (4k Long-Context):** Longformer Encoder-Decoder with sparse attention, processing up to 4,096 source tokens (hardware-capped from 16,384). Initialized from BART-large weights for a controlled architectural comparison.

- **BART-Oracle (1k Smart-Select):** An ablation baseline that uses global TF-IDF to rank and select the most salient 1,024 tokens from the *entire* document before passing them to BART-large, isolating the contribution of smarter truncation versus architectural context extension.

### 4.5 LoRA Fine-Tuning

Both BART-large and LED-large were fine-tuned using Low-Rank Adaptation (LoRA) [9], which freezes pre-trained weights $W_0$ and injects trainable low-rank decomposition matrices into the query and value projection modules (`q_proj`, `v_proj`):

$$W = W_0 + \Delta W = W_0 + AB^T \tag{4}$$

where $A \in \mathbb{R}^{d\times r}$ is Gaussian-initialized, $B \in \mathbb{R}^{r\times d}$ is zero-initialized (ensuring $\Delta W = 0$ at the start of training), and $r \ll d$ is the low rank. Training was conducted on a single NVIDIA A5000 24GB GPU using `bf16` mixed precision, AdamW optimizer, learning rate $3 \times 10^{-5}$, and weight decay 0.01 for 3 epochs. Table 3 summarizes the configurations. The lower LoRA rank for LED (4 vs. 8) accommodates the 4,096-token context window within the A5000's VRAM budget.

**Table 3** – LoRA hyperparameter configurations for BART and LED.

| Model | Context | Rank | Alpha | Dropout | Eff. Batch |
|---|---|---|---|---|---|
| BART-large | 1,024 | 8 | 32 | 0.05 | 16 ($4 \times 4$) |
| LED-large | 4,096* | 4 | 16 | 0.05 | 16 ($1 \times 16$) |

*Hardware-capped from the theoretical 16,384-token maximum.

### 4.6 Evaluation Metrics

All systems are evaluated on the held-out 1,500-sample test set using ROUGE [13] and BERTScore [24].

**ROUGE-N** measures $n$-gram overlap between a generated summary $\hat{s}$ and reference $r$:

$$\text{ROUGE-N} = \frac{\sum_{\text{gram}_n \in r} \text{Count}_{\text{match}}(\text{gram}_n)}{\sum_{\text{gram}_n \in r} \text{Count}(\text{gram}_n)} \tag{5}$$

We report ROUGE-1 ($n$=1) and ROUGE-2 ($n$=2). **ROUGE-L** uses the longest common subsequence (LCS) between $\hat{s}$ and $r$:

$$\text{ROUGE-L} = \frac{(1+\beta^2)\, R_{\text{lcs}}\, P_{\text{lcs}}}{R_{\text{lcs}} + \beta^2\, P_{\text{lcs}}} \tag{6}$$

where $R_{\text{lcs}} = \text{LCS}(\hat{s}, r)/|r|$, $P_{\text{lcs}} = \text{LCS}(\hat{s}, r)/|\hat{s}|$, and $\beta$ controls the precision–recall trade-off ($\beta \to \infty$ emphasises recall).

**BERTScore** [24] computes token-level cosine similarity between contextual embeddings of $\hat{s}$ and $r$. Precision $P_{\text{BERT}}$, recall $R_{\text{BERT}}$, and F1 are defined as:

$$R_{\text{BERT}} = \frac{1}{|r|} \sum_{r_j \in r} \max_{\hat{s}_i \in \hat{s}} \cos(\mathbf{r}_j,\, \hat{\mathbf{s}}_i), \quad P_{\text{BERT}} = \frac{1}{|\hat{s}|} \sum_{\hat{s}_i \in \hat{s}} \max_{r_j \in r} \cos(\hat{\mathbf{s}}_i,\, \mathbf{r}_j) \tag{7}$$

where $\mathbf{r}_j$ and $\hat{\mathbf{s}}_i$ are the contextual embedding vectors of tokens $r_j$ and $\hat{s}_i$ respectively.

Evaluation is conducted at two levels: (1) an **overall** aggregate across all 1,500 test articles, and (2) a **TIS-stratified** analysis within each bin. A Pearson correlation between per-sample TIS values

and per-sample ROUGE-2 gaps (LED minus BART) is computed to test the diagnostic hypothesis quantitatively. The BART-Oracle is evaluated using the oracle-selected input with BART's standard generation procedure.

## 5 Results & Discussion

### 5.1 Training Dynamics

Figure 3 shows the training and validation loss histories for both models. Both BART and LED exhibit smooth, monotonically decreasing loss curves over three epochs with no signs of divergence or late-stage overfitting. BART training loss decreases from approximately 3.4 to 2.5; LED descends from approximately 2.9 to 2.2. Notably, LED achieves a lower final training loss than BART, yet this advantage does not translate into higher ROUGE scores on the test set—underscoring the decoupling between cross-entropy optimization loss and n-gram overlap evaluation objectives.




**Figure 3** – Training and validation loss histories for BART-large and LED-large over three epochs. Both models converge stably. LED achieves a lower final training loss than BART yet does not outperform BART on ROUGE metrics, illustrating the gap between optimization and evaluation objectives.

### 5.2 Overall Evaluation Results

Table 4 presents aggregate metrics across all 1,500 test articles. BART-large achieves the highest scores on all ROUGE metrics: ROUGE-1 (0.3657), ROUGE-2 (0.1318), and ROUGE-L (0.2056). LED-large underperforms BART by 21.0% on ROUGE-1, 65.2% on ROUGE-2, and 24.7% on ROUGE-L. The overall mean ROUGE-2 gap (LED minus BART) is $-0.0520$.

BART-Oracle achieves ROUGE-2 of 0.1185, intermediate between BART and LED. Its recall (0.4550) substantially exceeds both BART (0.3613) and LED (0.3083), confirming that TF-IDF selection recovers more reference-relevant unigrams from throughout the document. However, its precision (0.3108) is the lowest of the three systems—below LED's 0.3511 and well below BART's 0.4154—and its ROUGE-L (0.1872) falls below BART's, indicating that scattered sentence selection disrupts the phrase-level continuity that ROUGE-L rewards.

### 5.3 TIS Distribution

Figure 4 shows the distribution of TIS scores across the 1,500 test articles. The distribution is left-skewed: the mean TIS is 0.4744 and the median is 0.5592, meaning the typical test article loses more than half of its salience-weighted reference-relevant content when processed by BART. The

**Table 4** – Overall evaluation results on the 1,500-sample test set.

| Model | ROUGE-1 | ROUGE-2 | ROUGE-L | Precision | Recall |
|---|---|---|---|---|---|
| BART-large (1k Baseline) | **0.3657** | **0.1318** | **0.2056** | **0.4154** | 0.3613 |
| LED-large (4k Long-Context) | 0.3022 | 0.0798 | 0.1648 | 0.3511 | 0.3083 |
| BART-Oracle (1k Smart-Select) | 0.3533 | 0.1185 | 0.1872 | 0.3108 | **0.4550** |

minimum TIS of 0.0000 indicates some articles are fully covered by the 1,024-token window, while the maximum of 0.9435 confirms near-total salient content loss in the most challenging documents. This median (55.92%) is distinct from the raw token discard rate of approximately 62% reported in the introduction: the former measures loss of *reference-relevant* content while the latter measures all tokens beyond position 1,024.

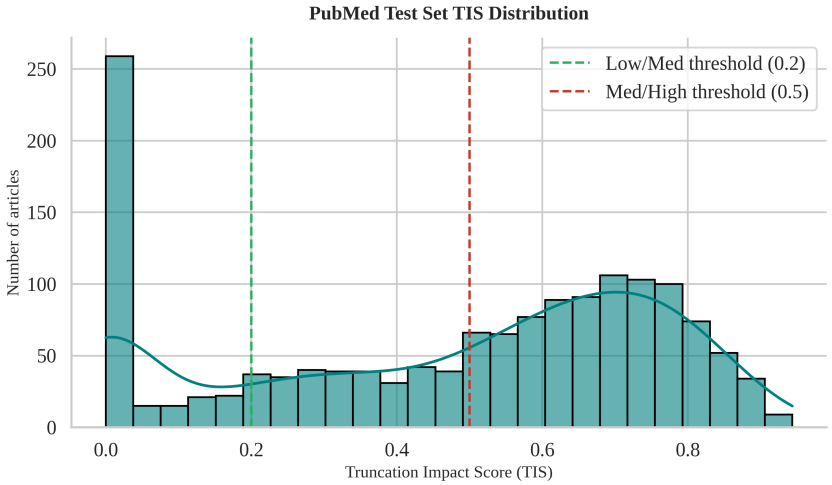



**Figure 4** – Distribution of TIS scores across the 1,500-sample test set. The median TIS of 0.5592 indicates that the typical test article suffers majority salience loss under a 1,024-token constraint. The heavy right tail confirms that severe truncation is common, not exceptional.

### 5.4 TIS-Stratified Evaluation

Table 5 and Figure 5 present the core empirical contribution: ROUGE-2 scores stratified by TIS difficulty bin.

**Table 5** – TIS-stratified ROUGE-2 results. The BART–LED gap narrows monotonically as truncation severity increases.

| TIS Bin | $n$ | Mean TIS | BART ROUGE-2 | LED ROUGE-2 | Gap (LED−BART) |
|---|---|---|---|---|---|
| Low ($< 0.2$) | 349 | 0.0353 | 0.1578 | 0.0948 | −0.0631 |
| Medium ($[0.2, 0.5)$) | 303 | 0.3581 | 0.1384 | 0.0806 | −0.0578 |
| High ($\geq 0.5$) | 848 | 0.6966 | 0.1188 | 0.0734 | −0.0454 |

Three findings emerge from the stratified analysis:

**Finding 1 — Monotonic performance decay for BART.** BART's ROUGE-2 declines from 0.1578 (Low TIS) to 0.1384 (Medium) to 0.1188 (High), a relative degradation of 24.7%. This confirms that as more salient content is discarded, BART's summarization quality decreases predictably—consistent with the diagnostic hypothesis of TIS.

**Finding 2 — Monotonic convergence of the BART–LED gap.** The ROUGE-2 gap narrows strictly from $-0.0631$ (Low) to $-0.0578$ (Medium) to $-0.0454$ (High). BART retains its advantage in all

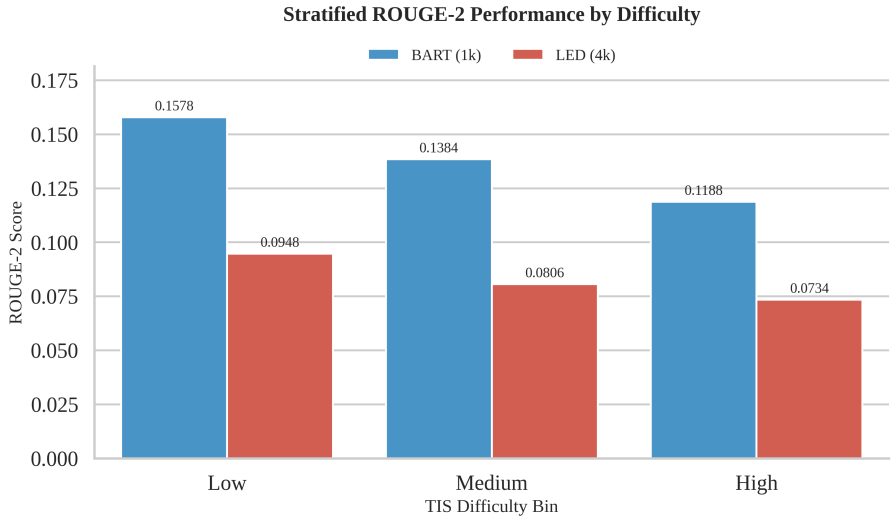



**Figure 5** – ROUGE-2 scores by TIS difficulty bin for BART-large and LED-large. The bars confirm monotonic performance decay for both models and a strictly narrowing BART–LED gap as TIS increases, validating TIS as a diagnostic signal for truncation-driven quality loss.

bins, but the advantage erodes by 28.1% from Low to High TIS. This convergence pattern is the primary empirical validation of TIS: it identifies the regime in which LED's long-context architecture provides meaningful relative benefit, even when it does not surpass BART in absolute terms under the current training configuration.

**Finding 3 — LED's partial compensation under severe truncation.** Even though LED does not close the gap entirely, its relative performance is best in the High-TIS regime—precisely the condition its architecture was designed for. LED ROUGE-2 declines from 0.0948 to 0.0734 across bins, a smaller slope than BART's, indicating partial resilience to increasing truncation pressure.

### 5.5 Correlation Analysis

Figure 6 plots per-sample TIS values against the per-sample ROUGE-2 gap (LED minus BART). The Pearson correlation is $r = 0.0789$, a statistically weak but positive relationship. This weak per-document correlation reflects high within-bin variance: many individual documents deviate substantially from the bin-level trend, and the per-sample ROUGE-2 gap is influenced by multiple confounders beyond TIS alone (e.g., abstract style, vocabulary specificity, generation randomness). Nevertheless, the bin-level analysis in Table 5 demonstrates a clear population-level monotonic signal that is meaningful despite this individual variability.

### 5.6 Oracle Ablation Analysis

Figure 7 provides a full metric-level comparison of all three systems. The oracle's high recall (0.4550) against its lower precision (0.3108) reveals a fundamental limitation of extractive selection as a surrogate for abstractive generation: TF-IDF ranking recovers reference-relevant unigrams effectively, but the selected sentences are not the sentences the author *chose to paraphrase* in the abstract. PubMed abstracts are authored as cohesive narratives rooted in the framing of the introduction, not as a distillation of the globally highest-salience sentences. The oracle's lower ROUGE-L further confirms that phrase-level fluency and sequential coherence are lost when sentences are reorganized by salience rank.

### 5.7 Discussion: The Counter-Intuitive BART Advantage

The finding that BART-large outperforms LED-large across all ROUGE metrics— despite a corpus where the median test article loses 55.92% of its salience-weighted content to truncation—is initially surprising. We propose the following explanation:

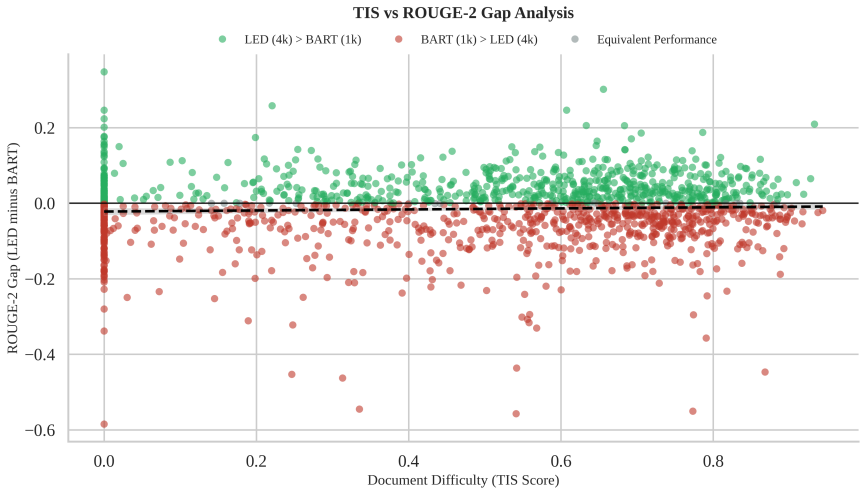



**Figure 6** – Per-sample scatter plot of TIS vs. ROUGE-2 gap (LED−BART). Pearson $r = 0.0789$. The weak linear correlation reflects high within-bin variance, yet the stratified bin analysis (Table 5) confirms a robust monotonic trend at the population level.

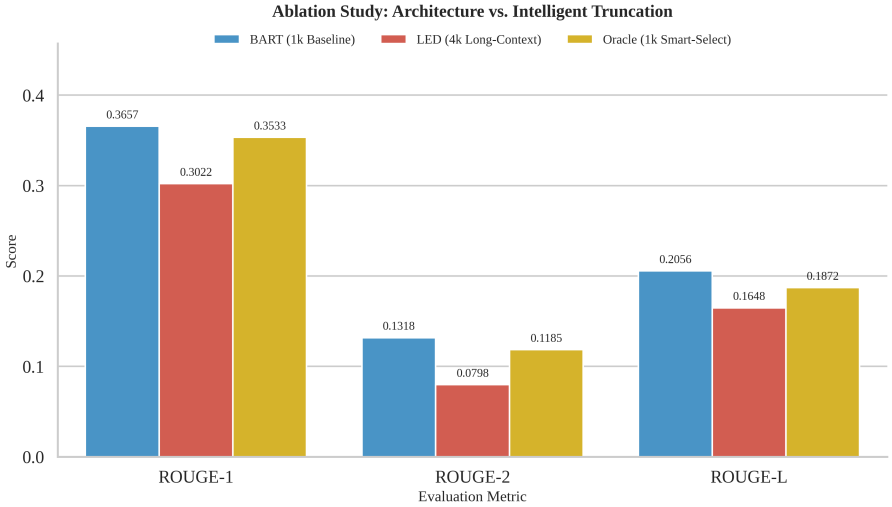



**Figure 7** – ROUGE-1, ROUGE-2, and ROUGE-L comparison across all three model systems. BART-large leads on all ROUGE metrics. The BART-Oracle achieves the highest recall but the lowest precision among the three systems (0.3108 vs. LED's 0.3511 and BART's 0.4154), reflecting the recall–precision trade-off of extractive sentence selection relative to the abstractive generation target.

**Positional lead bias.** PubMed abstracts tend to mirror the framing and terminology of the article's introduction and background sections—the very content appearing in the first 1,024 tokens. BART's simple right-side truncation therefore preserves the content most lexically aligned with the reference summary. The oracle result supports this: selecting globally most salient sentences via TF-IDF recovers more recall but *less* ROUGE-L, because those sentences are scattered throughout the document and are not the ones around which the abstract was actually written.

**Training alignment hypothesis.** LED's lower training loss versus BART's higher loss (yet better ROUGE) suggests LED may have over-optimized its cross-entropy objective in a way that does not generalize to n-gram evaluation. Future work should explore generation-quality metrics beyond ROUGE, such as BARTScore [21] and SummaC [11], which may reveal different rankings.

**Implications for TIS as a routing signal.** Despite BART's aggregate dominance, TIS-stratified evaluation shows that the gap shrinks significantly under high truncation pressure. For applications

where documents are known *a priori* to have high TIS—for example, methodology-heavy clinical trial reports—practitioners should consider routing to LED or similar long-context models, where the architectural advantage is most likely to manifest. Computing TIS requires only a TF-IDF lookup and unigram overlap computation, costing negligible inference time compared to a full model forward pass.

## 6 Conclusion

We have introduced and empirically validated the Truncation Impact Score (TIS), a model-agnostic, pre-inference diagnostic metric for quantifying salience-weighted content loss due to context window truncation in long-form biomedical summarization. Through a controlled study of 12,000 training and 1,500 test PubMed articles, we evaluated BART-large, LED-large (hardware-capped at 4,096 tokens from its theoretical 16,384-token maximum), and a BART-Oracle baseline fine-tuned via LoRA on a single NVIDIA A5000 24GB GPU.

Our primary findings are: (1) BART-large achieves superior overall ROUGE scores despite discarding approximately 62% of raw source tokens (and a median salience-weighted loss of 55.92% by TIS), an outcome attributed to positional lead bias in PubMed abstract authoring; (2) TIS reveals a monotonic convergence pattern—the BART–LED ROUGE-2 gap narrows strictly from $-0.0631$ (Low TIS) to $-0.0454$ (High TIS), validating TIS as a principled diagnostic for identifying the conditions under which long-context architectures offer the greatest relative benefit; and (3) the BART-Oracle's recall–precision trade-off demonstrates that intelligent extractive selection does not straightforwardly improve over naive truncation when evaluated against abstractive reference summaries authored from a lead-position framing perspective.

Future work should explore TIS in unsupervised settings where reference summaries are unavailable, using TextRank or lead-extractive approximations of salience. Extending the study to domains without strong positional lead bias—such as legal contracts or clinical discharge summaries—may reveal regimes where LED's architectural advantage manifests in absolute, not only relative, terms. Evaluation beyond ROUGE to faithfulness metrics such as SummaC and FactCC would provide a more comprehensive picture of the truncation–quality relationship.

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